

Absolute Market Unfairness - Measuring Market Inefficiency in Cryptocurrency Markets

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Abstract

This paper introduces *Absolute Market Unfairness (AMU)*, a model-free measure of the risk-free arbitrage profits that market makers can earn when quoted option prices deviate from put-call parity. We study market efficiency from a no-arbitrage perspective in cryptocurrency derivatives markets, where contract design differs substantially across exchanges and standard parity relations cannot be transferred without modification. On the theoretical side, we present a unified treatment of put-call parity for four contract classes: standard, inverse, standard quanto, and inverse quanto options. On the empirical side, we examine high-frequency data on BTCUSD and ETHUSD inverse options and matching futures from OKX and Deribit. Using synchronized bid-ask quotes and explicit trading-cost adjustments, we compute AMU in basis points as a directly comparable measure of investor-unfriendly pricing across platforms, underlyings, and market conditions. Our results show substantial cross-exchange differences in pricing efficiency: OKX appears more competitive overall than Deribit, while inefficiency is more pronounced in less liquid segments and during periods of market stress. The AMU measure provides a transparent summary statistic for comparing option market quality and may be useful for both market participants and regulators.

Keywords: cryptocurrency, put-call parity, market efficiency, market maker, bid-ask spread, inverse derivatives

JEL classification: G12, G14, G23

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1 Introduction

The rapidly expanding landscape of crypto trading platforms and exchanges, both in number and volume, offers investors a highly heterogeneous set of opportunities. Beyond cryptocurrencies as underlying assets, market participants trade a wide range of derivatives, including products whose underlyings are themselves derivatives. Although there is a large literature on consistent derivative pricing in mature markets and asset classes (e.g. Black & Scholes (1973), Sharma et al. (2024)), relatively little comparable research has focused on crypto exchanges (e.g., Alexander, Chen & Imeraj (2023), Lucic & Sepp (2024)). This gap is notable because extending established questions from mature markets to emerging asset classes is a natural research path. One reason for the gap may be strong cross-platform heterogeneity in contract design, which fragments market structure. While cryptocurrency futures are often liquid, options markets still feature a broad range of contract types, strikes, and maturities, many of which are thinly traded.

In this paper, we study quoted option prices from a no-arbitrage perspective using put-call parity (PCP), focusing on the liquid part of cryptocurrency derivatives markets where actionable pricing comparisons are meaningful. We introduce *Absolute Market Unfairness (AMU)*, a measure of the risk-free arbitrage profit implied by quoted bid-ask prices, and use it to compare investor-facing pricing quality across platforms.

Our contribution is threefold. First, we provide a unified PCP framework for standard, inverse, standard quanto, and inverse quanto contracts. Second, we adapt PCP relationships to platform-specific contract details and trading frictions. Third, we implement this framework on high-frequency BTCUSD and ETHUSD option and futures data from OKX and Deribit, and use AMU (in basis points) to compare market efficiency across exchanges, underlyings, and market states.

According to the Efficient Market Hypothesis (EMH) (Fama 1970), asset prices should reflect available information, so no participant can systematically earn excess returns from informational advantages alone. Evidence on crypto-market efficiency appears mixed at first sight ((Urquhart 2016, Le Tran & Leirvik 2020, Kristoufek 2018, Nan & Kaizoji 2019, Köchling et al. 2019, Vidal-Tomás et al. 2019, Wei 2018, Noda 2021)), but many studies agree that efficiency evolves

over time ((Urquhart 2016, Le Tran & Leirvik 2020, Köchling et al. 2019, Vidal-Tomás et al. 2019, Wei 2018)).

No-arbitrage pricing is a central dimension of market efficiency, because derivative valuation relies on price relationships that must hold in the absence of risk-free profit opportunities. A core relationship is put-call parity (PCP), which links European call and put prices to the underlying (or equivalently to a forward) at the same strike and maturity. Under ideal conditions (frictionless trading, liquidity, and competitive market making), PCP violations should be arbitrated away (Klemkosky & Resnick 1979). Testing PCP as an efficiency benchmark has a long tradition (e.g. (Klemkosky & Resnick 1979, Kamara & Miller 1995, El-Mekkaoui & Flood 1998, Cremers & Weinbaum 2010)), but crypto evidence is more recent (Alexander, Chen, Deng & Wang 2023, Zhou 2023, Felföldi-Szűcs et al. 2024, Johnson & Grigorev 2022). Across studies, violations are frequent when using mid or last prices, but become much smaller once high-frequency synchronization, transaction costs, and bid-ask spreads are incorporated (Wagner et al. 1996, El-Mekkaoui & Flood 1998, Nisbet 1992, Felföldi-Szűcs et al. 2024).

Our analysis is also tied to bid-ask spread economics. Spreads capture liquidity conditions, and prior work explains their size through adverse selection, inventory costs, and order-processing costs (Foucault et al. 2013), as well as market-maker pricing behavior (Glosten & Milgrom 1985, Brunnermeier & Pedersen 2009, Ding 1999). We extend this literature by quantifying the spread component that corresponds to risk-free arbitrage implied by PCP violations. This provides a direct lens on the maturity and competitiveness of crypto options markets.

Finally, from the viewpoint of our research question, the different crypto option contract designs are important. Alexander, Chen & Imeraj (2023) gives a summary of these contracts, as crypto exchanges differ fundamentally in contract design. The authors distinguish inverse options (e.g., Deribit) where collateral and payoff are defined in cryptocurrency; direct options, where crypto–stablecoin pairs are exchanged; quanto options, a new type of currency-protected options; and inverse-quanto option. Their main contribution is to show that crypto options create market incompleteness due to payoff currency mismatch, collateral denomination, embedded FX exposure and inability to perfectly hedge all risks. Their paper provides the first rigorous mathematical framework for pricing and hedging the main types of crypto exchange-traded

options—especially inverse, direct, and quanto options—within a Black–Scholes setting. Their results create a strong theoretical background for our research: since crypto derivative markets are heterogeneous and fragmented, original pricing relations cannot be naïvely ported from traditional markets, PCP must be adjusted for contract specification.

The remainder of the paper is structured as follows. Chapter 2 describes the analyzed platforms and the key descriptive statistics of the selected derivatives. Chapter 3 presents the PCP methodology for standard, quanto, inverse, and inverse-quanto option settings. Chapter 4 reports empirical results, followed by discussion and conclusion.

2 Dataset

2.1 Cryptocurrency platforms

There are several crypto trading platforms, and they differ in terms of product offerings, trading volumes, and market structure. Some of the most prominent platforms include Binance, OKX, Deribit, and BitMEX. Each platform offers a variety of derivative products, including standard futures, inverse futures, perpetual futures, and options with different underlying assets and payoff structures. Liquidity and trading volume can vary significantly across these platforms, which can affect market efficiency and the presence of arbitrage opportunities. Since our research focuses on options-market efficiency using put-call parity, we focus on platforms offering expiration (i.e., non-perpetual) futures where underlyings and expirations match the options we analyze. In this paper, we compare inverse options and inverse futures based on the same index (e.g., BTCUSDT) on OKX and Deribit. While options and futures on other underlyings are also traded, by far the most liquid options are those on the BTCUSD and ETHUSD indices, and thus we focus on these.

2.2 Data description

This paper is based on OKX and Deribit options data from Tardis, covering approximately one year from the beginning of 2025 to the end of February 2026. Table 1 summarizes how many put-call pairs we were able to study.

Tardis provides tick level intraday data for a wide range of cryptocurrency derivatives,

including options and futures contracts. The data includes detailed information on bid and ask prices, trade volumes, and timestamps for each contract. For our analysis, we subsampled the tick-level data on a 5 minute grid, which gives us intra-day resolution on the occurrence of arbitrage opportunities. Since most contracts only trade very few times within a day, this subsampling is not expected to distort the results.

	BTCUSD deribit	BTCUSD okex	ETHUSD deribit	ETHUSD okex
Avg num contracts	113	49	81	21
Avg num expirations	7	4	7	3
Avg num strikes	28	22	21	13
Avg call spread (bp)	32	20	49	137
Avg put spread (bp)	49	34	78	57
Num days in sample	73	72	73	72

Figure 1: Summary statistics of put-call pairs: number of contracts and liquidity measures.

3 Methodology

Typically, we think of options and futures contracts whose payoffs and strike parameters are defined in currency units (such as USD), as functions of differences in currency-denominated underlying prices. Such (so-called *linear*) options do trade on crypto platforms, but we found that *inverse* options, where payoffs are defined in terms of the underlying asset (e.g., BTC) rather than currency units, are more liquid and more frequently traded. Therefore, we focus on *inverse* options in our research, while also providing the theoretical framework for other option types (Alexander, Chen & Imeraj 2023, Bouchouev & Isakov 1997, Lucic & Sepp 2024).

In what follows we first provide a general introduction to the most frequently traded derivative products on crypto platforms, and then we define the modified versions of PCP equations for all product types. These theoretical results of our paper are generally valid PCP equations. Third, we add the transaction fees and all platform specific modifications to the PCP constraints, where the needed adjustments vary according to the given platform, and the validity of these PCP constraints is restricted to the given platform.

3.1 Underlying products and options on crypto derivative markets

The BTCUSD and ETHUSD indices serve as the primary underlyings for the options contracts analyzed in this study. These indices are constructed as composite reference prices, typically calculated as a weighted average of spot prices from multiple major cryptocurrency exchanges. The BTCUSD index reflects the market value of Bitcoin in US dollars, while the ETHUSD index represents the price of Ethereum in US dollars. Both indices are designed to minimize manipulation and provide a robust benchmark for settlement and pricing of derivative contracts. Options and futures contracts on platforms such as OKX and Deribit reference these indices to ensure consistency and transparency in contract valuation, even though the indices themselves are not directly tradable assets.

Beyond index prices, some platforms offer options and futures contracts settled on actual pair trade prices (e.g., BTC/USDT spot trades) or on perpetual futures. Perpetual futures are futures-like contracts without an expiry date, relying on a funding rate mechanism to anchor their price to the spot. While these products are increasingly popular and present unique settlement and pricing challenges, their payoff structures and lack of fixed expiration complicate the application of put-call parity and related arbitrage arguments. For this reason, we restrict our analysis in this paper to options and futures with well-defined expiry dates and matching underlyings, and do not consider contracts settled on actual pair trade prices or perpetual futures.

Although index prices are not directly tradable and lack their own bid-ask spreads, they provide a consistent and standardized reference for computing payoffs in derivative contracts. This allows for uniform settlement and valuation across products and platforms, regardless of the trading venue or asset denomination.

The fact that settlement is not actually based on the index introduces a layer of complexity to valuation, since settlement eventually happens either in the underlying asset of the index (e.g., BTC for BTCUSD) or in a stablecoin (e.g., USDT, the other asset in BTCUSD). Since payoffs to options and futures are defined through the same conversion from index value to coin units, put-call parity should still hold.

3.2 Put-call parity for crypto options

In the case of standard long forward contracts, the investor enters into an agreement at time $t = 0$ to purchase the quantity of the underlying asset specified in the contract at time $t = T$ (the expiry date) at the forward price F_0 . At the moment of expiry, when the investor buys at the forward price F_0 , the value of the position is:

$$(S_T - F_0) \quad (1)$$

However, for futures, those are exchange-traded contracts, profits and losses do not need to be realized at expiry, as mark-to-market settlement is continuous, meaning that there is a settlement at the end of each trading day or, in the case of cryptocurrency futures, usually at the end of every 8 hours, as the trading on these exchanges are often 24/7.

The value of a futures contract f_t at an arbitrary time t , and at maturity T within the interval $(0, T)$ is:

$$f_t = (F_t - F_0), \quad \text{final payoff: } F_T = (F_T - F_0) \quad (2)$$

This will be added to or deducted from the variable margin of the investor.

Inverse futures keep the well-defined expiry date of standard futures, but change the denomination of the contract. While the contract itself is quoted in USD, margin, profit and loss, and settlement are defined in the underlying asset (e.g., a cryptocurrency such as Bitcoin), which makes the original linear value and final payoff shown in Equation 2 nonlinear, as follows:

$$f_t = \left(\frac{1}{F_0} - \frac{1}{F_t} \right), \quad \text{final payoff: } f_T = \left(\frac{1}{F_0} - \frac{1}{F_T} \right) \quad (3)$$

.

Trading platforms offer not only inverse futures but also perpetual futures, which have gained importance recently, but are not part of our research since PCP constraints cannot be adapted to them in a closed form. In contrast to the previous products, a perpetual future is a futures-like contract starting at $t = 0$ but with no expiration date (T), which means it never

settles automatically, and investors trade it continuously as they would a spot instrument. The expiry date and convergence-to-spot-at-expiry are replaced by a funding-rate mechanism to anchor futures prices to spot prices. At each funding interval (e.g., every 8 hours), if $F_1 > S_1$, investors with long futures positions pay $(F_1 - S_1)$ to short investors; if $F_1 < S_1$, short investors pay $(S_1 - F_1)$. For further insight into perpetual futures, see He et al. (2022), who provide both a theoretical pricing framework and an empirical analysis based on a constant-arbitrage strategy demonstrating that real-world perpetual futures often trade at significant deviations from these fundamentals in cryptocurrency markets. While Ackerer et al. (2025) offers a general no-arbitrage pricing framework, it applies to both linear and inverse perpetual futures.

Option contracts may differ not only by their underlying (LU), but also by their payoff structure or their payoff currency. As Alexander, Chen & Imeraj (2023) provides a step-by-step guide for understanding inverse, linear, and inverse quanto options, we simply refer to their summarised results in Table 1.

Theoretical no-transaction-cost formulas for the put-call-parity relationships for the four cases are shown in Table 2. The differences between the four versions arise entirely from what the “effective underlying” is and whether a quanto adjustment modifies its domestic-currency forward price. The put-call parity relationships are derived under the assumption of no dividends from the underlying, and that the discount factor is given by $D(t, T) = e^{-r(T-t)}$ where r is the risk-free interest rate.

3.3 *Definition of the MU arbitrage*

MU stands for *Market Unfairness*. It is a way to characterize whether the bid-ask spread in a certain market is unfairly large. But what is unfair? We define the *MU arbitrage* as the risk-free profit that can be generated by market makers by quoting on the bid/ask of the put/call contract, and then completing the full arbitrage cycle. The full arbitrage cycle consists of trading the put and call options, and then trading the underlying asset (or forward contract) to complete the cycle. If the bid-ask spread is too wide, then it creates a risk-free profit opportunity for market makers, which is a sign of market inefficiency. The *MU arbitrage* quantifies this inefficiency in terms of basis points of the underlying asset’s value.

Option Type	Underlying Product	LU Formula	LU Currency	LC Payoff at T	LC Pay-off Currency
Standard Option	Spot BTC/USD	S_T	USD per BTC	$\max(S_T - K, 0)$	USD
Inverse Option	Inverse BTC/USD Contract	$\frac{1}{S_T}$	BTC per USD	$\max\left(\frac{1}{S_T} - \frac{1}{K}, 0\right)$	BTC
Standard Quanto Option	BTC/USD settled in BTC	S_T	USD per BTC	$\frac{1}{S_0} \max(S_T - K, 0)$	BTC
Inverse Quanto Option	Inverse BTC/USD settled in USD	$\frac{1}{S_T}$	BTC per USD	$S_0 \max\left(\frac{1}{S_T} - \frac{1}{K}, 0\right)$	USD

Table 1: Comparison of Standard, Inverse, and Quanto Crypto Options (In the table, the USD is seen as Domestic Currency and BTC represents the Cryptocurrency.)

Case	Put-call parity relationship	Notes
Standard option	$C_t - P_t = D(t, T)(F_t - K) = S_t - KD(t, T)$	Payoff and premium in USD.
Inverse option	$C_t^{\text{inv}} - P_t^{\text{inv}} = D(t, T) \left(\frac{1}{F_t} - \frac{1}{K} \right)$	Payoff and premium in coin units (e.g. BTC).
Standard quanto option	$C_t^{\text{q}} - P_t^{\text{q}} = \frac{1}{S_0} D(t, T)(F_t - K)$	Linear PCP scaled by fixed conversion factor $1/S_0$.
Inverse quanto option	$C_t^{\text{iq}} - P_t^{\text{iq}} = S_0 D(t, T) \left(\frac{1}{F_t} - \frac{1}{K} \right)$	Inverse PCP scaled by fixed conversion factor S_0 .

Table 2: Put-call parity relationships for the four contract cases. Here $D(t, T) = e^{-r(T-t)}$ denotes the discount factor. We assume no dividends from the underlying.

An important feature of our inefficiency definition is that it is completely model-free. We do not need to impose any assumptions on volatility dynamics, market impact, inventory risk, or risk premia in order to quantify the arbitrage-based deviation from put-call parity. Trading costs are incorporated separately and transparently as observable adjustments to the arbitrage bounds. For this reason, including such costs does not distort the cross-exchange comparison of inefficiency; it only shifts the feasible arbitrage region in a directly measurable and comparable way.

3.4 Comparing platforms using the AMU measure

The *AMU* measure is defined as the average of the positive *MU arbitrage* opportunities observed over a certain period of time, expressed in basis points of the underlying asset's value. This measure allows us to compare the efficiency of different trading platforms and to track the evolution of market efficiency over time. A higher *AMU* indicates a greater degree of market inefficiency, as it reflects more frequent or larger risk-free profit opportunities for market makers due to wide bid-ask spreads that deviate from put-call parity.

3.5 MU arbitrage paths

The mechanism for the theoretical arbitrage is as follows: The market maker observes that *if* she joined the top of the market on the bid/ask of the put/call contract (these are the 4 arbitrage paths), then when filled a risk-free profit could be realized by completing the full arbitrage cycle. The *fwd_call* path corresponds to the forward direction, with the market maker joining the bid of the call leg. The *bck_put* path corresponds to the backward direction, with the market maker joining the bid of the put leg, and so on.

From a market maker's perspective, these are very weak opportunities, since profit requires another market participant to cross the market maker's passive order. However, keeping an order in the book has no direct cost, and market makers are often required to maintain a reasonable market for a substantial share of time and contracts, so posting such quotes can still be rational. In a well-functioning market, such opportunities would disappear, since market makers have incentives to improve bid-ask prices in competition with each other.

4 Results

In the following we present AMU results in basis point units so that ETH and BTC results can be easily compared. AMU in bp is the average dollar AMU per observed put-call pair divided by the value of the underlying, accounting properly for the contract size.

Even though AMU can also be expressed in dollar terms (arbitrage opportunities have a definite dollar size), cross-platform comparisons should not penalize more liquid markets where smaller proportional AMU can still correspond to larger dollar amounts. Therefore, in what follows, we focus on AMU measured in basis points.

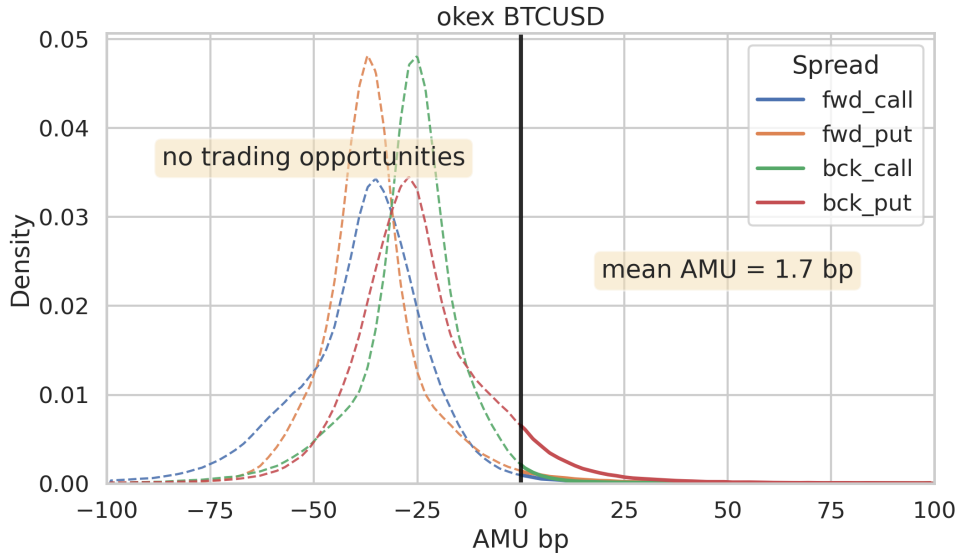


Figure 2: Distribution of the four versions of the AMU arbitrage on OKX for BTCUSD option pairs. Only the positive tail corresponds to actual tradable opportunities. 30 bp round trip trading costs were assumed.

Figures 2, 3, 4, and 5 show the distribution of the four versions of the AMU arbitrage across exchange and underlying combinations. In all cases, only the positive tail corresponds to actual tradable opportunities, while the negative tail corresponds to non-tradable situations. The mass of the distribution remains concentrated on the negative side, showing that most of the time quoting is not profitable as a path to risk-free profit via the PCP-breaking relationship.

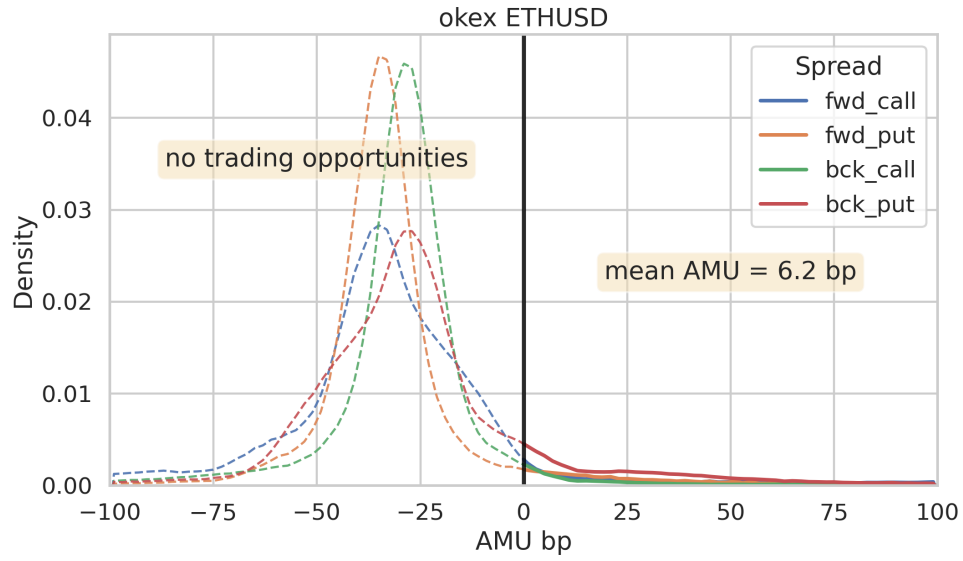


Figure 3: Distribution of the four versions of the AMU arbitrage on OKX for ETHUSD option pairs. Only the positive tail corresponds to actual tradable opportunities. 30 bp round trip trading costs were assumed.

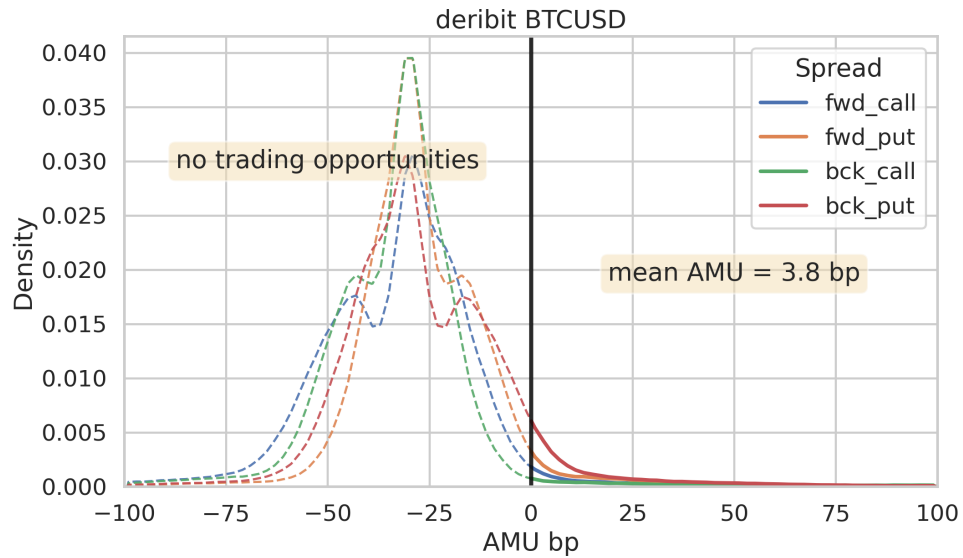


Figure 4: Distribution of the four versions of the AMU arbitrage on Deribit for BTCUSD option pairs. Only the positive tail corresponds to actual tradable opportunities. 30 bp round trip trading costs were assumed.

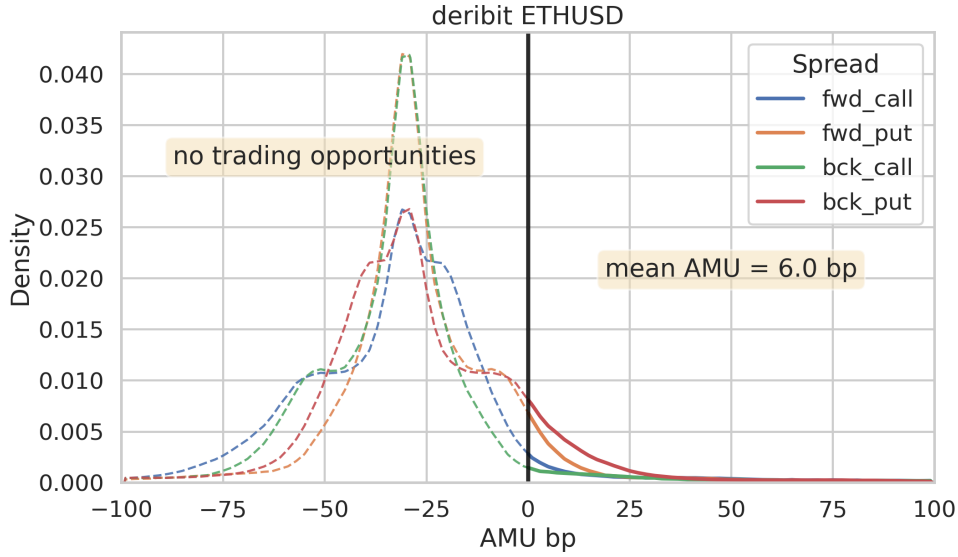


Figure 5: Distribution of the four versions of the AMU arbitrage on Deribit for ETHUSD option pairs. Only the positive tail corresponds to actual tradable opportunities. 30 bp round trip trading costs were assumed.

4.1 Trading costs and the AMU arbitrage

Trading costs are crucial in evaluating the actual value of the AMU arbitrage. While we carefully addressed bid-ask spreads, trading fees must also be considered. We assume a simple fee structure where the total cost of an arbitrage round trip is 30 basis points, a reasonable value across the analyzed exchanges. More important, fees shift the AMU distribution leftward without distorting cross-exchange comparisons of inefficiency; they only shift the feasible arbitrage region in a directly measurable and comparable way. After accounting for trading fees, tradable AMU opportunities are those above 30 basis points.

The time evolution of the AMU arbitrage

In figure 6 we show the time evolution of the AMU across exchanges. When superimposed with the BTCUSD spot value, it is notable how the AMU becomes much higher during times of market stress, namely times of sudden BTC price drops.

The bottom pane of figure 6 shows how often the AMU arbitrage actually sets up out of all times when we have valid put and call markets. The notable weekly structure of the number of ticks has to do with a data artifact pertaining to Fridays in the raw Tardis data.

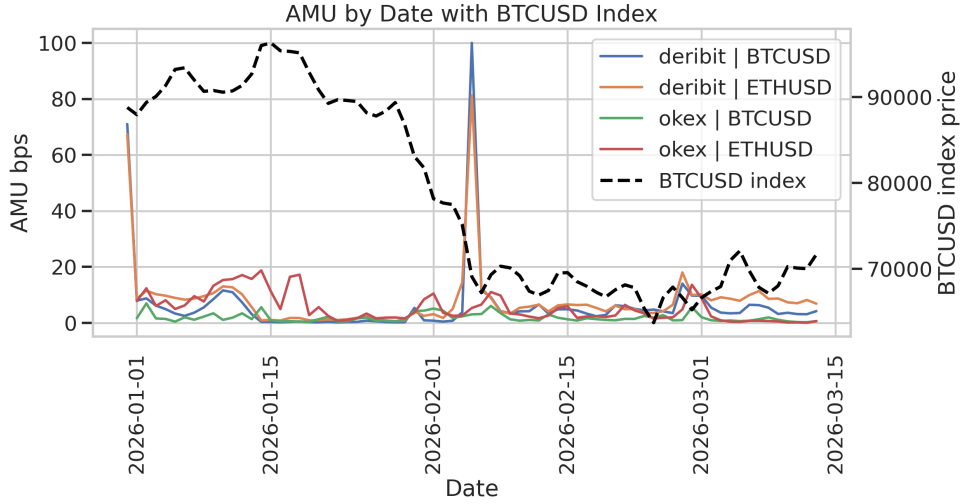


Figure 6: Time evolution of AMU (in basis points) across exchanges for BTC and ETH options

4.2 AMU by Time-to-Expiration

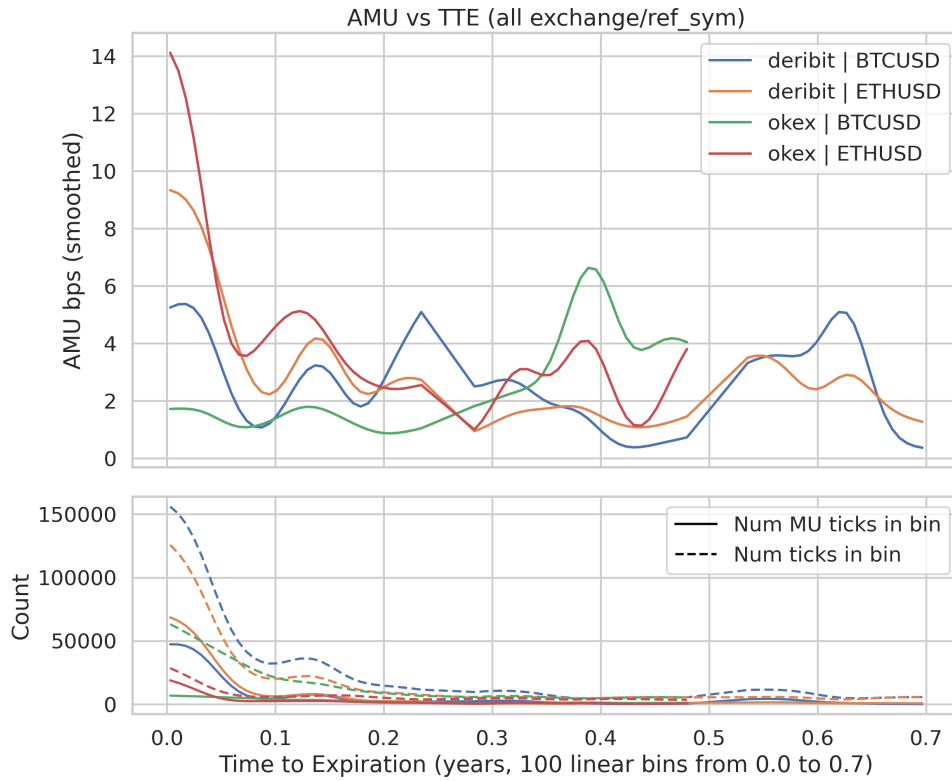


Figure 7: AMU (in basis points) across exchanges stratified by time-to-expiration (TTE) for BTC and ETH options.

TTE refers to the time to expiration of the option pairs observed, measured in years. As shown in figure 7, the AMU is worse for farther dated options. However, there are very few observations available for longer dated contracts, most of the data is for options expiring in two

weeks or less.

4.3 AMU by Relative Strike

Relative strike is the conventional measure K/F , where K is the strike price and F is the forward price of the underlying. Relative strike = 1 corresponds to a put-call pair where both contracts are at the money. For relative strike < 1, the call is in the money and the put is out of the money, while for relative strike > 1, the put is in the money and the call is out of the money. As shown in figure 8, the AMU is worse for out-of-the-money options, and is best for at-the-money options. This is consistent with the fact that liquidity is usually the best for at-the-money options, and the worst for out-of-the-money options, which makes it more likely to observe PCP breaking.

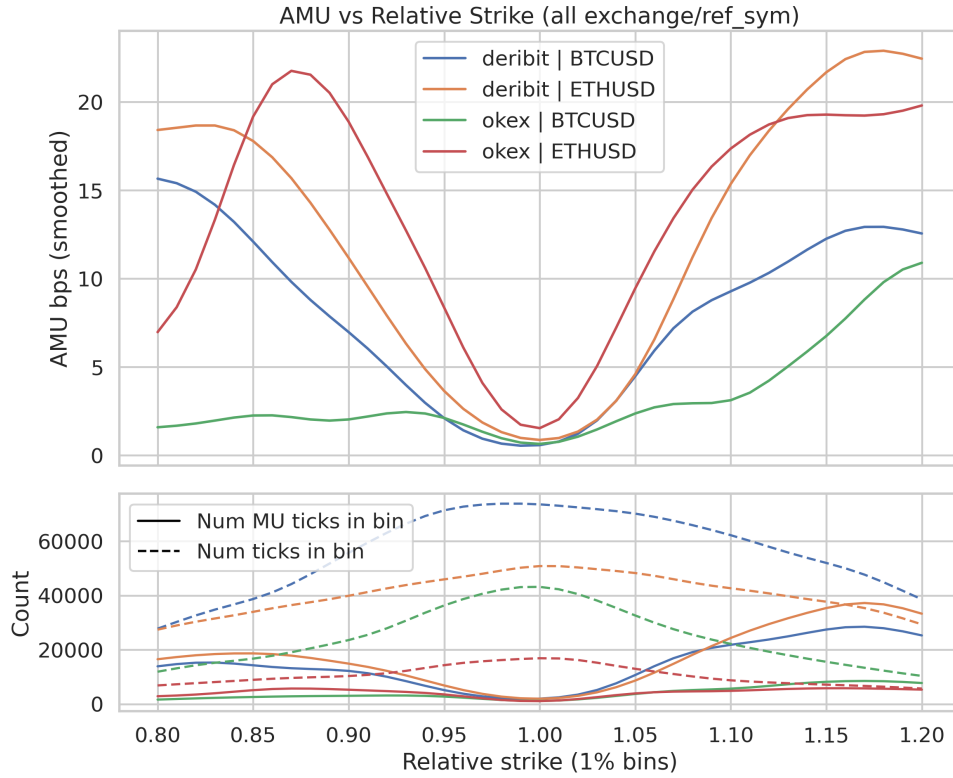


Figure 8: AMU (in basis points) across exchanges stratified by relative strike (moneyness) for BTC and ETH options

	AMU(bps)	MU occurs	Num Observations
BTCUSD deribit	3.76	587,255	2,317,423
BTCUSD okex	1.66	165,152	1,018,224
ETHUSD deribit	5.98	705,279	1,666,474
ETHUSD okex	6.18	173,137	435,179

Figure 9: Summary of AMU arbitrage results across exchanges and underlyings.

4.4 Discussion

4.5 Platform Specific Contracts and Put-Call Parities

4.5.1 OKX futures and options

For notation, we use S_t to denote the BTCUSD index at time t , and $S_t^{(U)}$ the BTCUSDT index at time t . These index prices are determined as averages of multiple exchanges' settlement prices. U_t is the USDUSDT exchange rate at time t , and superscripts (U) denote quantities related to the BTCUSDT index (as opposed to the BTCUSD index). In this market, futures and options payouts depend on the value of S_t , which is a published index value rather than a directly traded price. It therefore has no bid-ask spread.

For OKX futures, we describe the crypto derivatives in a world where the numeraire is USDT, the stablecoin tied to the US dollar (OKX 2024a).

4.5.2 Crypto-margined (BTCUSD) Futures

Crypto-margined BTCUSD futures are margined and settled in BTC, with a face value of 100 USD and a contract multiplier of 1. The forward contract corresponding to the position “long 1 BTC-USD-250627 future”, has the following payoff at maturity T :

$$\text{Value}_T = 100 \times \left(\frac{1}{F} - \frac{1}{S_T} \right) \quad (\text{in BTC}) \quad (4)$$

where T is 2025-06-27, and F is the futures price at inception. Futures prices are quoted as a value of the BTCUSD index.

USDT-margined BTCUSDT futures are settled and margined in stablecoin-margined (USDT), with a face value of 0.01 BTC and a contract multiplier of 1. The forward contract corresponding

to the position “long 1 BTC-USDT-250627 future”, has the following payoff at maturity T :

$$\text{Value}_T = 0.01 \times (S_T^{(U)} - F^{(U)}) \quad (\text{in USDT}) \quad (5)$$

where T is 2025-06-27, and $F^{(U)}$ is the futures price at inception. Futures prices are quoted as a value of the BTCUSDT index.

Analogous futures exist for ETHUSD and ETHUSDT.OKX (2024a).

4.5.3 *Crypto-margined (BTCUSD) Options*

Regarding OKX options, the platform offers European-style options settled in BTC — not in futures or stablecoins (OKX 2024b). The multiplier is 0.01. The underlying index is BTCUSD. The payout of one contract call BTC-USD-250627-80000-C option is

$$\text{Value}_T = 0.01 \times \frac{\max\{0, S_T - K\}}{S_T} \quad (\text{in BTC}) \quad (6)$$

$$\text{Value}_T = 0.01 \times \max\{0, S_T - K\} \quad (\text{in USD, theoretically}) \quad (7)$$

$$\text{Value}_T = 0.01 \times \max\{0, S_T - K\} \times U_T \quad (\text{in USDT, after the sale of the BTC received}) \quad (8)$$

$$(9)$$

where $K = 80000$ is the strike price, and T is 2025-06-27 and U_T is the USDUSDT exchange rate at maturity. The actual payout is in BTC, but in the put-call-parity arbitrage path, a BTC-USDT conversion will be necessary. If you are long one such call option, the premium you pay at inception is $0.01 \times c_t$ (in BTC), which defines the scaling of c_t . The put option is analogous in the usual way.

Option premia are quoted in BTC. Analogous options exist for ETHUSD and ETHUSDT.OKX (2024b).

4.5.4 *Put-Call Parity in OKX*

From here on we assume $U_T = 1 \pm \sigma_U\%$, with $\sigma_U \approx 0.001\%$ (Binance 2024). With this assumption, we can write down the put-call parity relationship between OKX options and the

OKX USDT-margined futures with matching strikes and expirations:

$$(c_t - p_t) \times S_t = (F_t^{(U)} - K) \exp(-r(T - t)) \quad (10)$$

Note the conversion of the option premia from BTC to USD on the left-hand side. The r is the USD interest rate over the period $[t, T]$. Also, note that we did not need the "dividend yield" for BTC, because the futures price is available. As ?? splits after considering bid-ask prices, 10 behaves similarly as seen in 11 and 12:

$$(F_t^{(U)} - K)e^{-r(T-t)} - (c_{ask,t} - p_{bid,t})S_{ask,t} \geq 0 \quad (11)$$

$$-(F_t^{(U)} - K)e^{-r(T-t)} + (c_{bid,t} - p_{ask,t})S_{bid,t} \geq 0 \quad (12)$$

4.6 Computing AMU in the data

In Felföldi-Szűcs et al. (2024) we described the *forward* and *backward* arbitrage paths, and their dependence on various cost components. The general concept was to define the forward and backward arbitrage profit as 13 and 14 when violating simple, textbook-version options.

$$\text{Profit}_{\text{forward}}^{\text{Simple}} = S_{ask} - C_{bid} + P_{ask} - Ke^{-rT} \quad (13)$$

$$\text{Profit}_{\text{backward}}^{\text{Simple}} = Ke^{-rT} - S_{bid} + C_{ask} - P_{bid} \quad (14)$$

In this paper, we apply the same approach to contracts across several crypto-derivatives markets. The dominant cost component in Felföldi-Szűcs et al. (2024) was the options bid-ask spread, so we focus on that, with the approximate additive commission discussed above.

We'll compute PCP breaking in units of USDT, and then convert to basis points of the underlying by dividing by the value of the underlying corresponding to one contract.

For the *forward* path, we sell the call for c_t^{bid} BTC, then sell the BTC for S_t^{bid} USDT, buy the BTC for S_t^{ask} USDT, buy put for p_t^{ask} BTC, and long the future at $F_t^{(U),\text{ask}}$. The strike is given in

the USD-denominated index, so that has to be "sold" at U_t^{bid} .

For a transaction size of 1 contract, the profit of the forward path is given by

$$\text{Profit}_{\text{forward}} = \left[c_t^{\text{bid}} \times S_t^{\text{bid}} - p_t^{\text{ask}} \times S_t^{\text{ask}} - (F_t^{(U),\text{ask}} - K) \exp(-r(T-t)) U_t^{\text{bid}} - \text{comm} \right] \times 0.01 \quad (15)$$

a commission rate of 30 bps on the notional of each trade would give us

$$\text{comm} = 0.003 \text{ BTC} = 0.003 \times S_t^{\text{bid}} \text{ USDT} \quad (16)$$

Finally, for a transaction size of 1 contract, the profit of the backward path is given by

$$\text{Profit}_{\text{backward}} = \left[p_t^{\text{bid}} S_t^{\text{bid}} - c_t^{\text{ask}} S_t^{\text{ask}} + (F_t^{(U),\text{bid}} - K) e^{-r(T-t)} U_t^{\text{ask}} - \text{comm} \right] \times 0.01 \quad (17)$$

The final formula for AMU as used in our empirical analysis is then given by

$$\text{AMU (bps)} = \left[\frac{\text{Profit}^{\text{forward or backward}}}{0.5 \times (S^{\text{bid}} + S^{\text{ask}}) \times 0.01} \right]^+ \times 10000 \quad (18)$$

Note that AMU does not depend on the multiplier.

4.7 OKX ETHUSD, and Deribit BTCUSD and ETHUSD futures and options

Inverse options, futures, and options on OKX and Deribit have identical payoff structures, even though contract multipliers can differ. It follows that, without any change, the formula also applies to other inverse contracts on both exchanges.

5 Discussion

General liquidity considerations can explain why ETHUSD options are less efficient than BTCUSD options, and why OTM/ITM options are less efficient than ATM options. Longer-dated options are more efficient than shorter-dated ones. However, there are nontrivial observations:

We can conclude overall that OKX is a fairer market than Deribit (see Figure 9). It is important to note that the main driver of this difference is that spreads on OKX are much more reasonable away from ATM options. OKX option markets also remain more reasonable during periods of stress.

5.1 Effect of trading costs on the AMU arbitrage

What should be done if costs differ across exchanges? For example, if we assume 60 bps for OKX and 30 bps for Deribit, then the AMU distribution on OKX would shift leftward by 30 bps, which would make OKX AMU appear even better (we would observe a much smaller part of the MU tail). To make the comparison fair, we assume the same reasonable cost level, because we should not penalize an exchange in our evaluation for having lower costs.

Data Availability Statement

The data that support the findings of this study are available through Tardis (*Tardis.dev – Main* 2026). Historical datasets for the first day of each month are openly accessible without an API key, as documented in the Tardis dataset description *Tardis.dev – Main* (2026), *Tardis.dev – Documentation* (2026). Access to data for all other days of the month requires an active subscription.

Supplemental Online Material

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6 Appendix

6.1 Call Parity for Foreign Currency Options

Forward price of foreign currency: $F_{t,T} = S_t \exp((r - r_{BTC})(T - t))$, where r is the domestic interest rate (USD), and r_{BTC} is the foreign interest rate (BTC). The put-call parity for foreign currency options is

$$c_t - p_t = S_t \exp(-r_{BTC}(T - t)) - K \exp(-r(T - t)) \quad (19)$$

Rearranging, we have

$$(c_t - p_t) = (F_{t,T} - K) \exp(-r(T - t)) \quad (20)$$

Hull (2017)